Introduction to Systems of Linear Equations

29 September 2025

Study Guide

This week, we will be learning about:

- linear equations, linear systems, and their solutions;
- augmented matrices of linear systems;
- the elementary row operations, Gaussian and Gauss-Jordan elimination;
- matrices in row echelon form and reduced row-echelon form.

You may find it helpful to list down many examples *and* counterexamples of the various objects we encounter in this week's class. Moreover, we talk about the following key result:

Proposition 1. *Every system of linear equations has zero, one, or infinitely many solutions. There are no other possibilities.*

Definitions

Definition 2. A **linear equation** in n **variables** $x_1, x_2, \ldots, x_n$ is an equation which can be expresed in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, a_2, \ldots, a_n$ and b are constants, and the coefficients a_i's are not all zero. In the special case where $b = 0$, we have a **homogeneous linear equation**.

Definition 3. A finite set of linear equations is called a **system of linear equations**, or a **linear system**. The variables in linear systems are called **unknowns**. A linear system can be represented by an array of numbers:

$$\left\{\begin{array}{ccccccc} a_{11}x_1 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}\right. \equiv \left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right).$$

This array is called the **augmented matrix** of the linear system.

Definition 4. A **solution** of a linear system in n unknowns $x_1, \ldots, x_n$ is a set of numbers $s_1, \ldots, s_n$ for which the substitution

$$x_1 = s_1, \quad x_2 = s_2, \quad \ldots, \quad x_n = s_n$$

makes each equation in the system a true statement. If a linear system has *at least one solution*, we say that it is **consistent**; otherwise, it is **inconsistent**—that is, the linear system has no solutions. If a linear system only has one solution, we say that the solution is **unique**.

Definition 5. A matrix is said to be in **row-echelon form** if:

1. The first non-zero element in each row, called the **leading entry**, is in a column to the right of the leading entry in the previous row.
2. Rows consisting of only zeros are grouped at the bottom of the matrix.

A matrix is said to be in **reduced row-echelon form** if it is in row-echelon form and it has the following additional properties:

1. All of the matrices leading entries are 1's.
2. The leading entry in each row is the only non-zero entry in its column.

Exercises

1. (a) Find a linear equation in x and y that has a general solution

$$x = 1 + 2t, \quad y = t, \quad t \in \mathbb{R}.$$

   (b) Show that

$$x = t, \quad y = \frac{t}{2} - \frac{1}{2}, \quad t \in \mathbb{R},$$

   is also a general solution for the equation in (a).

2. (a) Find a linear equation in x, y, and z that has a general solution

$$x = 3 - 4s + t, \quad y = s, \quad z = t, \quad s, t \in \mathbb{R}.$$

   (b) Express a general solution for the equation in (a) in two other ways.

   (c) Write down a linear system of two different nonzero linear equations such that the system has the same general solution as in (a).

3. For each question below, answer "yes" or "no," justifying your answers.

   (a) Does an inconsistent linear system with more unknowns than equations exist?

   (b) Does a linear system which has only one solution, but more equations than unknowns, exist?

   (c) Does a linear system which has infinitely many solutions, but more equations than unknowns, exist?

4. Find a series of elementary row operations that bring the augmented matrix $\boldsymbol{A}$ to the augmented matrix $\boldsymbol{B}$, where

$$\boldsymbol{A} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right) \quad \text{and} \quad \boldsymbol{B} = \left(\begin{array}{ccc|c} 3 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 9 & 9 \end{array}\right).$$